MAKE THINGS HAPPY !

IoTextra Octal3

Digital I/O with Latching Relay Module

The **IoTextra Octal3** (Rev. 3.02) is an eight-channel digital I/O module, containing four digital input and four latching relays.
The maximum digital input signal level is 36VDC.
Module has four **KEMET EC2-5SNU** bistable (latching) relays. Contact form — **DPDT (2 Form C)**. Relay switching is performed via the **Texas Instruments DRV8837C** H-bridge driver, which provides polarity reversal on the coil with a short pulse. Relays retain their state without power applied to the coil.
All digital channels have individual galvanic isolation with a dielectric strength of 3750 Vrms.
Up-to-date technical information about the module is available on **GitHub**.

SOFTWARE POWER:

The **IoTextra Octal3** module offers native support for popular IoT platforms:

Node-RED | Home Assistant | TASMOTA | Blynk | IoTflow

This hardware compatibility unlocks the full potential of the **IoTflow platform**. It enables you to:

- Create custom **Node-RED** nodes effortlessly.
- Integrate via **MQTT** with third-party global platforms like **Blynk**.
- Scale your project from a single node to a complex industrial network.

Integrate with **Tasmota** for reliable **MQTT**-based telemetry and remote control — support for **IoTextra** modules, including **IoTextra Octal3**, is described **here**.

The module provides indication of the digital input channel status and relay state.

The **IoTextra Octal3** module requires both the **HOST** connector (for power supply, digital input signals **IN1–IN4**, and **nSLEEP** control) and the **I²C** bus (for relay output control via the **TCA9534** I/O expander).
Up to 16 modules can be connected to one I²C bus, provided that the **nSLEEP** signal (shared across all modules or individual per module) is supplied to each module — this signal is required to enable the **DRV8837C** relay drivers.

This module is designed for systems requiring reliable state retention across power cycles, combined with high-speed digital input monitoring and galvanic isolation of switched loads.

APPLICATIONS:

⚡ **IIoT — Control Cabinet Management**
Digital inputs monitor the status of circuit breakers and voltage presence; latching relays hold external contactors or disconnect switches in position through power outages with zero coil energy consumption.
Recommended software: Node-RED, IoTflow

📡 **Telecommunications — Bypass and Line Switching**
Digital inputs monitor line status and equipment health; latching relays provide physical bypass switching of backup lines, remaining in position indefinitely without holding current — critical for equipment running continuously for years.
Recommended software: Tasmota

🌱 **Smart Farming — Irrigation Control**
Digital inputs track water level sensors and valve limit switches; latching relays activate irrigation valves with a short pulse and hold them open with zero energy consumption, enabling precise scheduling in solar-powered installations.
Recommended software: Node-RED, IoTflow

🏠 **Smart Home — Blinds and Gate Control**
Inputs monitor position sensors and manual buttons; latching relays control blinds, gates, or ventilation dampers, retaining their last position after a power cut — no unexpected movement on power restoration.
Recommended software: Home Assistant

FEATURES:

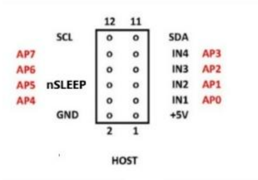
- Compatibility with major microcontrollers
- Module power supply is 5VDC, which feeds a 3V3 regulator for powering the logic section
- Includes a 3V3 Indicator LED
- Protection against reverse power supply polarity
- 4 independent, optically isolated digital input signals with TVS protection on the inputs
- Insulation strength for input signals (opto-coupler characteristic) from the module's logic section is 3750Vrms
- Digital input voltage is up to 36VDC:
  - 0 to 2 V – opto-coupler closed, logical “1” on opto-coupler output
  - 4 to 36V – opto-coupler open, logical “0” on opto-coupler output
- 4 bistable (latching) **DPDT (2 Form C)** type relays (**KEMET EC2-5SNU**)
- Switching Voltage: 250VAC / 220VDC
- Max. switching current: 2A
- Coil resistance [±10%] (at 20°C 68°F) -- 250 Ω
- Relay switching is performed via **TI** DRV8837C H-bridge driver; state change requires a short pulse per EC2-5SNU Datasheet
- All four drivers share a common **nSLEEP** control signal (**AP5** on **HOST** connector)
- Relays retain their state without power applied to the coil; relay state must be stored in firmware (EEPROM/flash)
- The module features LED indicators for the status of digital input channels and relay state (LED lights up when the opto-coupler on the input is open / relay is energized)
- Reading the input signal states is done using signals **IN1-IN4** (**AP0-AP3**) through the **HOST** connector
- Relay outputs are controlled via the **I²C** bus through the TCA9534 I/O expander, which drives the **DRV8837C** H-bridge drivers. The common nSLEEP signal (**AP**5 on **HOST**) enables or disables all relay drivers simultaneously
- The I/O expander used in the module is the TCA9534 or a compatible chip
- The expander's **I²C** address (**A2-A0**) is set using DIP microswitch:

TCA9534	0	1	0	0	A2	A1	A0	x
TCA9534A	0	1	1	1	A2	A1	A0	x

- Connection to the module via the **I²C** bus is done through [Qwiic©]() connectors or through pins 11 (**SDA**) and 12 (**SCL**)
- Transient suppression and electrostatic discharge protection of signals on **Qwiic©** connectors is done using a TVS diode assembly (ESD protection)
- Module size is 47x56 mm. The module has mounting holes allowing it to be installed in the base module or in a Raspberry Pi

HOST CONNECTOR:

The pinouts for these **HOST** connectors are shown in the image below:



The **HOST** connector is used differently depending on how the microcontroller interacts with the **IoTextra Octal3** module:

1) ***Standalone Use:*** In this mode, the input signal states and control of the output signals are available via the **HOST** connector and the **I²C** bus using the I/O expander.
2) ***Smart Use:*** In this mode, an **IoTsmart** microcontroller module is vertically inserted into the **HOST** connector. Reading the input signal states, control of the output signals and the 5V module power supply are done through the **HOST** connector. Power is supplied from the **IoTsmart** module. Below is a photo of the **IoTextra Octal3** module with the **IoTsmart** module:

IoTextra Octal3 with **IoTsmart ESP32-S3**

IoTextra Octal3 with **IoTsmart RP2350A**

IoTextra Octal3 with **IoTsmart RP2040**

3) ***Mezzanine Use:*** In this mode, the **IoTextra Octal3** module is installed in the base module, the state of the input signals and the output signal control are available through the **HOST** connector. Photo of the **IoTextra Octal3** module installed in **IoTbase** modules below.

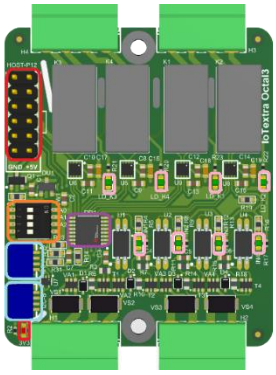
IoTbase PICO with **PICO2** and **IoTextra Octal3** modules

IoTbase NANO with **Waveshare ESP32-S3-NANO** and **IoTextra Octal3** modules

COMPONENT LAYOUT:

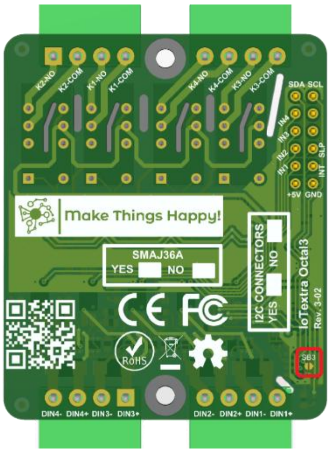
There are three versions of the **IoTextra Octal3** module, and accordingly, each with a different **top-side** component layout:

It is possible to obtain input signal status via the **HOST** connector (highlighted in red).
The **Qwiic©** connectors are highlighted in light blue.
The I/O expander is highlighted in purple.
The DIP microswitch for setting the I/O expander's address on the **I²C** bus is highlighted in orange. The “ON” position of the microswitch corresponds to a “0” value in the respective address bit (A0, A1, and A2) for the I/O expander. The “OFF” position corresponds to a “1” value.
There is a 3V3 power indicator. The power indicator are highlighted in red in the **top-side** image.
The I/O signal indicators are shown in pink in the figure.



JUMPERS:

The image below shows the component and marking layout on the **bottom-side** of the **IoTextra Octal3** module:

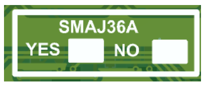


The **bottom-side** of the **IoTextra Octal3** module has a jumper **SB3**, that allows powering devices connected through the **Qwiic©** connector. It is highlighted in red. By default, the jumper is absent.

CONFIGURATION TABLES:

The bottom-side of the module provides confiration information about **I²C** connectors:

TVS diodes **SMAJ36A** installed or not



EXTERNAL SIGNAL CONNECTION:

Removable terminal blocks with a 3.5 mm pitch are recommended for connecting external input signals to the **H1-H2** connectors on the module. The pin layout is marked on the **bottom-side** of the module.

The relays are connected to **H3- H4** connector.

ACCESSORIES:

The following accessories may be required for using the module:

- A set of four removable terminal blocks with a 3.5mm pitch for terminal blocks **H1-H4**
- A set of two standoffs and four screws for mounting the module into the **IoTbase** series base module.
- Cable for the **Qwiic©** connector